How CNN Explainer teaches a convolution

A design review of an interactive explanation, from one photograph to the calculation behind one coloured square.

A reader can recognise a pizza long before they can read a neural network. CNN Explainer uses that head start. The photograph, the scores on the right and the patterns in between belong to the same prediction. When a reader opens one pattern, the page owes them an explanation of something they have already seen.

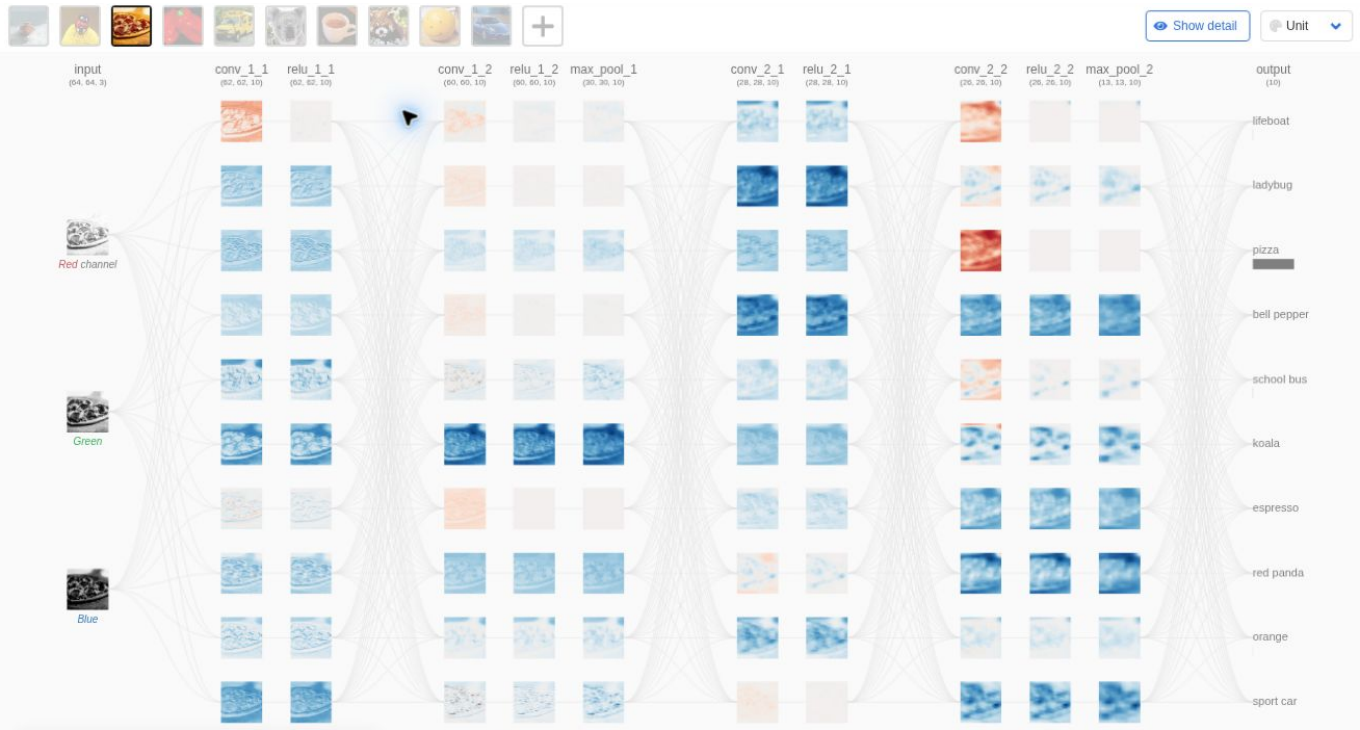


Figure 1. The pizza example on the live page. The selected photograph is at the top left, the three colour channels are on the left and the longest output bar is beside “pizza”. The small coloured squares are maps of numerical responses. [1]

The design choice worth copying

Keep one example visible as the question changes. First ask what the system produced. Then ask which inputs contributed to a selected result. Only then show the rule that combines them. The middle question gives the arithmetic a job to do.

The page offers this sequence through clicks. Visitors can also read the article, inspect another layer or change the image. That freedom suits a reader who already has a question. A first-time visitor has to discover where to begin.

Review method. Desktop walkthrough of the pizza example, inspection of the accompanying article and controls, and review of the authors’ published user study. Observations, interpretations and proposed adaptations are distinguished below.

Three questions, one photograph

1. What did the model choose?

Selecting pizza updates the input channels, the coloured maps and the class scores together. Pizza has the longest bar. The architecture stays in place, so a reader comparing it with the espresso example can see what depends on the photograph.

2. Where did this map come from?

Click the top square in the first convolution column. The surrounding network fades and the selected path opens out. Each colour channel gets its own small grid of learned weights, called a kernel. Three intermediate maps feed a plus sign. The model adds their values at matching positions and adds a learned offset, or bias.

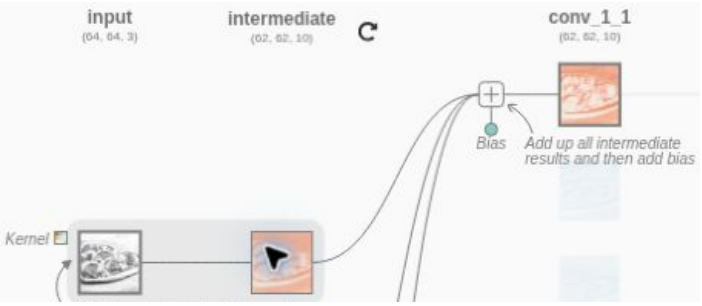


Figure 2. Detail of the expanded path. The crop shows the top channel and the junction where contributions from all three channels meet. [1]

3. How did one position get its value?

Click the intermediate map beside the first input channel. The panel puts the image patch on the left, nine pixel-by-weight products in the centre and the corresponding result on the right. Moving to a different position changes the patch being sampled. The same nine weights are used again.

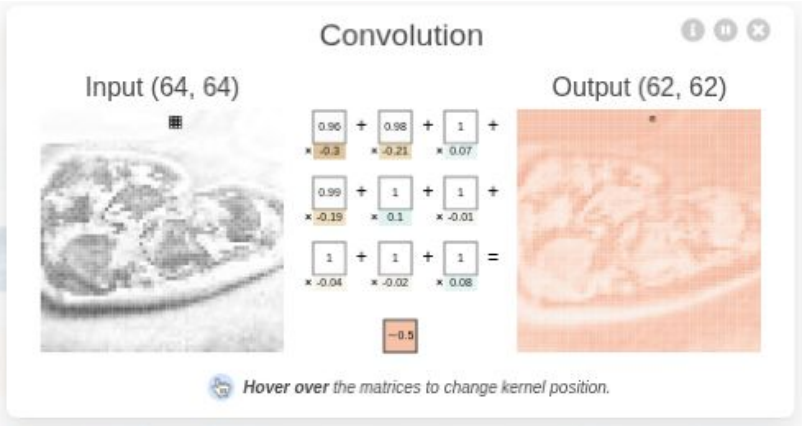


Figure 3. A single channel in the formula panel. A 3 × 3 window moves across a 64 × 64 input, producing a 62 × 62 intermediate map. [1]

Why the middle view matters. The number in Figure 3 explains one channel's contribution at one location. The junction in Figure 2 shows what still has to happen to produce the selected map. Without that junction, a reader could reasonably mistake one channel's arithmetic for the entire calculation.

What each design choice buys

The page gives text, colour, motion and controls different jobs. Each earns its place when it helps the reader answer a particular question.

A familiar starting point	The photograph and class names provide a question in ordinary language before the reader encounters “activation” or “kernel”. A recognisable outline sometimes persists in early maps. That gives the learner a visual reference while the underlying quantity changes. [1]
Detail in place	Opening a convolution makes room around the selected map and fades unrelated connections. The transition lets the eye follow what moved. A separate diagram would require the reader to work out which part of the original network it represented. [1, 2]
Two uses of motion	Moving layers apart explains where the reader is. Moving a kernel across an image explains what the model repeats. The first animation can be brief. The second needs pause and inspection because its subject is the changing relationship between a patch and a result. [1, 2]
Words beside the action	The expanded view labels the sum and bias where the three paths meet. The article then directs readers to a specific first-layer map and walks through its three inputs. A layer heading links to its article section. This combination serves both readers who start by exploring and readers who need instructions first. [1]
A different view for a different rule	ReLU is a point-by-point rule. A negative value becomes zero and a positive value passes through. Max pooling compares neighbours and keeps the largest. The former calls for a before-and-after pair, the latter for a highlighted group and its winning value. Reusing one generic animation would hide that distinction. [1]
A separate place to experiment	The article’s grid lets the reader vary input size, padding, kernel size and stride. Its displayed starting case uses a 5 × 5 input and a 2 × 2 window, producing a 4 × 4 output at stride 1 with no padding. Counting four legal positions per side makes the output size intelligible. This experiment is separate from the trained image classifier. [1]

Where colour helps, and where it misleads

The page distinguishes activations from learned weights using different palettes and provides numerical scales. The scale selector also changes what a colour can be compared with. Two equally blue squares can represent different values under different ranges. For a comparison task, keep a shared scale visible or print the two values alongside it.

How much confidence is warranted?

What the evaluation supports

The authors observed 16 university students using the tool. Fourteen were postgraduate students. Participants received a five-minute CNN introduction, a three-minute tool demonstration and a feature checklist. They explored while thinking aloud, then completed a questionnaire and interview. [2, sections 8.1–8.3]

9 of 16	Participants who commented favourably on transitions between the network and its mathematical detail. Their comments support the interpretation that keeping the network in view helped them connect levels. [2]
6.88 / 7	Mean enjoyment rating. Ease of understanding averaged 5.94 / 7. These are answers to different questions, and both are self-reports. A reader can enjoy following an animation while still needing help to explain it. [2, figure 8]

The study supports a claim about the experience of these learners during a supported session. Learning gains over a static explanation, recall a week later and unaided first use by a general audience remain open questions. Some beginners wanted more guidance. [2]

Where a replica could go wrong

The first screen can ask too much.

A novice meets many small maps, faint links and abbreviated names at once. The worked route sits in the article below. Offer one visible invitation tied to an object, such as “Choose the first coloured square to see where it came from.” Let experienced readers skip it.

A vivid map can invite too strong a conclusion.

A coloured region shows a numerical response. Its resemblance to a pizza edge can suggest a hypothesis about what the map responds to. Establishing that the edge caused the final choice would require a different test. Likewise, a normalised class score alone cannot establish how often the classifier is correct.

Hovering and colour can become prerequisites.

The formula panel explicitly asks the reader to hover, and some values and controls are small. For another audience, provide a selectable cell, keyboard steps, text values and a still view of the same calculation. Keyboard, assistive-technology and mobile usability were outside this walkthrough.

The experiment has a boundary.

The grid can demonstrate exactly how a larger stride changes output dimensions. It cannot establish a change in classification accuracy. Label what a control actually changes and separate measured effects from claims that would require training and testing a model.

How to build and test the same idea

Start with one question your audience already asks and one result they can recognise. Work backwards until you can show the smallest step that result depends on. Build that one route before adding a catalogue of controls.

Worked adaptation for an evidence explorer

Suppose an illustrative dashboard says that 6 of 8 firms filed in a given year (75%). A reader wants to know which firms count. The numbers below are invented to show the interaction.

1. Show the result	Display “6 of 8 firms filed (75%)”, the year and the definition of a firm. Make “Which firms count?” the first action beside the figure.
2. Open its inputs	Keep the result visible and reveal eight firm rows. Mark the six with a qualifying filing. Selecting a row opens its dated source record and the inclusion rule. This is the equivalent of expanding one map into its contributing channels.
3. Change one fact	In a labelled practice copy, move one of the six firms’ only qualifying filings outside the year. Ask for a prediction before revealing 5 of 8 (62.5%). Highlight the changed row and the numerator. The denominator stays eight because the cohort was fixed in advance.
4. Return and explain	Close the detail with the same row selected. Ask why the percentage fell and what evidence would justify counting the firm again. The reader must connect the rule, the record and the headline. A generic “learn more” link cannot supply that connection on its own.

Test whether the reader can use the rule

Give first-time readers an unfamiliar case without a walkthrough. Record whether they can find the first action unaided, predict the changed result and identify the record and rule that explain it. Score each separately. Someone who finds the detail but predicts the wrong denominator has exposed a conceptual problem that a click count would miss.

If choosing between formats, give a second group the same examples and wording as annotated stills. Compare accuracy on a fresh case and ask for the reasoning. Measure enjoyment separately. The comparison tests whether interaction earns its extra design and development cost.

Primary sources

[1] Polo Club. CNN Explainer, live visualisation and accompanying article. Figures 1–3 are cropped captures of the pizza example from the desktop walkthrough.

[2] Wang, Z. J. et al. (2021). CNN Explainer: Learning Convolutional Neural Networks with Interactive Visualization. IEEE Transactions on Visualization and Computer Graphics. See sections 6 and 8, especially figure 8.